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Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2025

Code: 105504		***Time: 03 Hours***
Semester: V	***Subject: Software Engineering***	***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Which model is most suitable for high-risk projects?
- (i) Spiral Model
- (ii) Waterfall Model
- (iii) Prototyping
- (iv) Incremental Model

(b) Which model is also known as the linear sequential model?
- (i) Agile Model
- (ii) Spiral Model
- (iii) Waterfall Model
- (iv) RAD Model

(c) Which of the following is the primary goal of requirement analysis?
- (i) Coding the software
- (ii) Understanding and documenting user needs and requirements
- (iii) Testing the software
- (iv) Designing the software architecture

(d) Requirement elicitation techniques include
- (i) Interviews
- (ii) Observation
- (iii) Questionnaires
- (iv) All of these

(e) Cohesion refers to
- (i) Inter-module dependency
- (ii) Internal strength of a module
- (iii) Testing quality
- (iv) User interaction

(f) Which UML diagram represents dynamic behavior?
- (i) Sequence diagram
- (ii) Class diagram
- (iii) Component diagram
- (iv) Package diagram

(g) Function Point Analysis is used to measure
- (i) The number of lines of code in a software project
- (ii) The functionality delivered to the user
- (iii) The performance of the code
- (iv) The user interface complexity

(h) COCOMO is used for
- (i) Testing
- (ii) Cost estimation
- (iii) Requirement analysis
- (iv) Debugging

(i) White-box testing is also called
- (i) Functional testing
- (ii) Acceptance testing
- (iii) Structural testing
- (iv) Black-box testing

(j) Which testing technique uses cause-effect relationships?
- (i) Cause-effect graphing
- (ii) Mutation testing
- (iii) Path testing
- (iv) Unit testing

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain the software crisis. Discuss its causes and how Software Engineering addresses the crisis.

(b) Write down the differences between Prototype Model and Incremental Model with advantages and limitations.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Explain requirement validation and requirement management processes.

(b) Discuss the characteristics of a good Software Requirement Specification and its components.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain coupling and cohesion with their types and significance in software design.

(b) What are the goals of good software design? Explain with examples.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Compare LOC and Function Point metrics with advantages and disadvantages.

(b) Explain Halstead's Software Science metrics and their applications. How does it help in assessing software quality?

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Explain black-box and white-box testing techniques with examples.

(b) Discuss Boundary Value Analysis and Equivalence Class Testing.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Discuss ISO 9000 quality standards and SEI CMM levels.

(b) Explain the role of documentation in software maintenance and quality assurance.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Write down the differences between Functional and Non-functional requirements with suitable examples.

(b) Discuss Brooks “No Silver Bullet” concept and explain why software productivity remains challenging.

Q.9 Solve both questions : **[7 x 2 = 14]**

(a) Explain Spiral Model with diagram and discuss how risk management is incorporated in each cycle.

(b) Define the COCOMO model. Also, explain the types of COCOMO model and their benefits.

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